



PROJECT REPORT

Cloud Computing



**GLOBAL PROFESSIONAL
INTERNSHIP (GPI)**

Problem Statement

 INDIVIDUAL PROJECT  120 HOUR



Cloud Computing



Cloud Counselage

Mr. John Doe is an employee at Cloud Counselage. He works in the IT department. He works on the Cloud technology. Cloud counselage has many Business division and as well many domains hosted on the cloud. Mr. John is hosted a website on AWS EC2 instance along with MongoDB database. But taking backups of the database manually is time-consuming.

So Mr. John wants to automate taking database Backup continuously without getting bogged down in different backup and failure scenarios in MongoDB database systems. He wants to utilize the time he spends on the backup task to do more productive work.

So, we want you guys to create a POC [proof of Concept] on the effective Automatic MongoDB backup Method. So that Mr. John will no longer need to spend hours on that process.

We want you guys to come up with a solution and create POC [Proof of Concept] for the same.

Note:
You are free to use any technology, tool and programming language of your choice. We won't be providing any AWS account credentials; you can create and use your own free tier AWS account for PoC purpose.

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Created On:	09-08-2026	Approved On:	DD-MMM-YYYY

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General Instructions for using the Live Project Report Template

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- The **text between inequality (< >)** is to be replaced by relevant text
- Please **remove the yellow highlight on the Text** between the inequality (< >). This is done to help you notice the text to be changed/replaced
- The text in *italics* highlighted in grey is just for reference and should be removed after adding the relevant text

1 PROJECT DETAILS

Project Name	Automated Database Backup Infrastructure		
Project Sponsor	Mr. John Doe / Cloud Counselage Pvt. Ltd.		
Project Manager			
Start Date	07-08-2026	Completion Date	09-08-2026

2 SUMMARY

This project delivers an automatic backup system for Mr. John Doe's web application. Manual backups take too much time and can have human errors. This system saves data automatically every day. The long-term benefit is total data security. If the server crashes, we can recover the data quickly.

3 INTRODUCTION

3.1 Background

Mr. John Doe runs a web infrastructure that uses a MongoDB database. This database stores important company data. Taking backups manually every day was difficult and risky. Therefore, we needed an automated solution to protect the data without human effort.

3.2 Stakeholders

- **Mr. John Doe:** Client and business owner.
- **Cloud Counselage Pvt. Ltd.:** Internship provider and evaluator.
- **Cloud Engineer:** the intern who built the solution.

3.3 Objectives

- To connect with the MongoDB database on an Ubuntu Server.
 - To write a simple Bash Script that fetches database data.
 - To set a timer (Cron Job) that runs the script automatically every night at 12:00 AM.
 - To compress the data into a ZIP file and save it in a safe folder called `backups_data`.
-

4 METHODOLOGY

These conventions are all about the positions of line breaks, how many characters should go on a line, and everything in between.

4.1 Considerations & Assumption

- We assume the Ubuntu Server is always running and connected to the internet.
- We consider that the local disk storage has enough space to hold the daily ZIP files.

4.2 Approach

We used a structured automation approach. Instead of manual work, we used Linux tools like `mongodump` for fetching data and `crontab` for automatic scheduling. This keeps the system highly reliable

4.3 Activities

- Studied the project requirements.
- Created the `backups_data` folder on the Ubuntu server.
- Wrote and tested the Bash Script code.
- Configured the Cron Job scheduler for 12:00 AM.
- Verified that the ZIP files are creating properly.

5 TARGETTED V/S ACHIEVED OUTPUT

- **Targeted Output:** An automatic script that runs daily at midnight, creates a ZIP backup, and saves it locally.
 - **Achieved Output:** The system is fully ready. The Bash script runs exactly at 12:00 AM, creates the ZIP file, and moves it to local storage. There are no deviations or errors.
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6 CONCLUSION

This automated backup system successfully protects Mr. John Doe's business data from accidental loss. It improves system efficiency. For future scope, we can upgrade this system to upload the ZIP files directly to cloud storage like **AWS S3** instead of local storage.

7 APPENDICES

7.1 Appendix A – Project Architecture Diagram

